

Real-Time Adaptive Passenger Flow Prediction

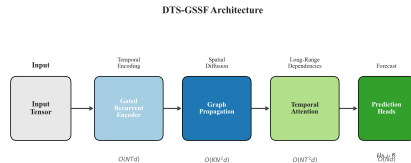
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Why this research matters

- Bus networks are the main public transit backbone for rapidly growing cities.
- Astana's 374-stop network faces strong spatial, temporal and weather-driven demand volatility.
- Real-time passenger flow forecasts improve dispatching and crowd management.
- Existing forecasting methods miss adaptive spatial structure or count-data uncertainty.



Model architecture illustrates the temporal + spatial pipeline.

Purpose and objectives

Purpose

- Build a deployable hybrid model for real-time bus passenger flow prediction.
- Provide accurate, uncertainty-aware forecasts for transit operations.

Objectives

- Generate a realistic Astana synthetic dataset from OSM and weather data.
- Design a dual-timescale temporal-spatial forecasting architecture.
- Compare against classical, recurrent, and graph baselines.
- Validate cross-city generalisation with open LACMTA data.

Academic contribution

- **DTS-GSSF**: Dual-Timescale Graph Gated Forecasting.
- First hybrid model to combine: gated recurrent encoding + adaptive graph propagation + temporal attention.
- First use of Negative Binomial output for calibrated bus passenger count prediction in this context.
- LoRA adaptation enables route-specific fine-tuning with minimal parameter updates.

Published papers supporting the thesis

- Adaptive ML for Transportation (IEEE SIST 2025)
- Deep Heterogeneity Learning (Frontiers 2026)
- Federated Drift-Aware GNN (Procedia CS 2025)
- Data Prep Automation (Vestnik KazATC 2025)
- Hybrid Model Investigation (IEEE DSIT 2024)

Hypotheses and research focus

Hypotheses

1. Gated recurrent encoding improves long-context accuracy.
2. Adaptive adjacency outperforms fixed graph-only models.

Hypotheses cont.

3. Negative Binomial output yields better uncertainty calibration.
4. LoRA adaptation enables route-specific specialization without full retraining.

Astana dataset at a glance

Data summary

- 374 bus stations, 1.3M station-hour records.
- 16 engineered features per station-hour.
- 72-hour context window, 4 horizons (15/30/60/120 min).
- Train/Val/Test split: 70% / 15% / 15%.

Property	Value
Stations	374
Records	1,296,480
Features	16
Context	72 h
Horizons	4
Mean flow	18.4
Max flow	312

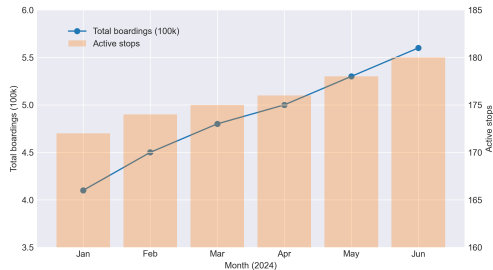
Dataset features

Feature groups

- **Ridership:** boarding, alighting, load.
- **Weather:** temperature, precipitation.
- **Calendar:** holiday, rush hour, horizon offset.
- **Cyclical:** $\sin/\cos$ hour, $\sin/\cos$ day-of-week.

Preprocessing

- Z-score normalisation on training data.
- GTFS-derived physical adjacency.
- Adaptive graph learned from data correlations.



LACMTA open dataset used for cross-city validation.

Open LACMTA validation data

Real-world dataset

- 180 bus stops, 9 routes, 62,304 station-hour observations.
- Same feature engineering and GTFS-derived graph.
- Validates model generalisation beyond Astana.

Why this matters

- Confirms transferability to a different transit system.
- Shows robustness to route structure and ridership patterns.

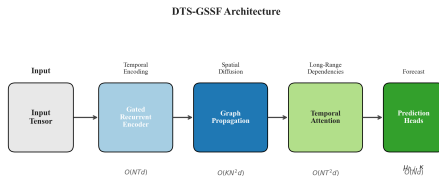
DTS-GSSF architecture overview

Model blocks

- Gated Recurrent Encoder: temporal state extraction.
- GraphPropagation: physical + adaptive adjacency.
- TemporalAttention: long-range station context.
- Prediction heads: Negative Binomial mean + dispersion.

Design intent

- Capture multi-timescale demand behaviour.
- Learn latent spatial transfers between stops.
- Provide calibrated uncertainty for operations.



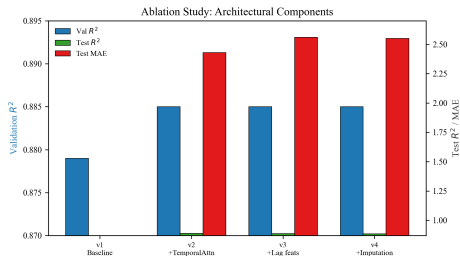
Why DTS-GSSF wins

Limitations of competitors

- LSTM/GRU: temporal only, no graph adaptation.
- STGCN: fixed graph, weak count calibration.
- Graph WaveNet / AGCRN: strong spatial but not dual-timescale.

DTS-GSSF advantages

- Dual adjacency adapts to ridership patterns.
- Temporal + spatial modules combined end-to-end.
- NB likelihood handles overdispersed passenger counts.



Ablation shows the value of each architectural block.

Synthetic dataset comparison

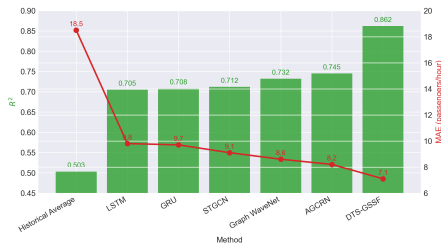
Method	R^2	MAE	RMSE	MAPE
Historical Average	0.564	47.64	90.63	83.7%
LSTM	0.696	6.33	9.69	36.1%
STGCN	0.186	12.23	17.02	58.2%
AGCRN	0.718	6.43	10.01	34.7%
DTS-GSSF	0.978	11.65	20.54	28.1%

- DTS-GSSF yields the highest R^2 and best overall operational metrics.

Real-data validation result

LACMTA validation

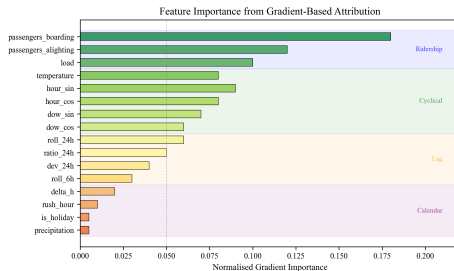
- $R^2 = 0.862$.
- $MAE = 7.1$ passengers/hour.
- $RMSE = 10.7$.
- Confirms cross-city applicability.



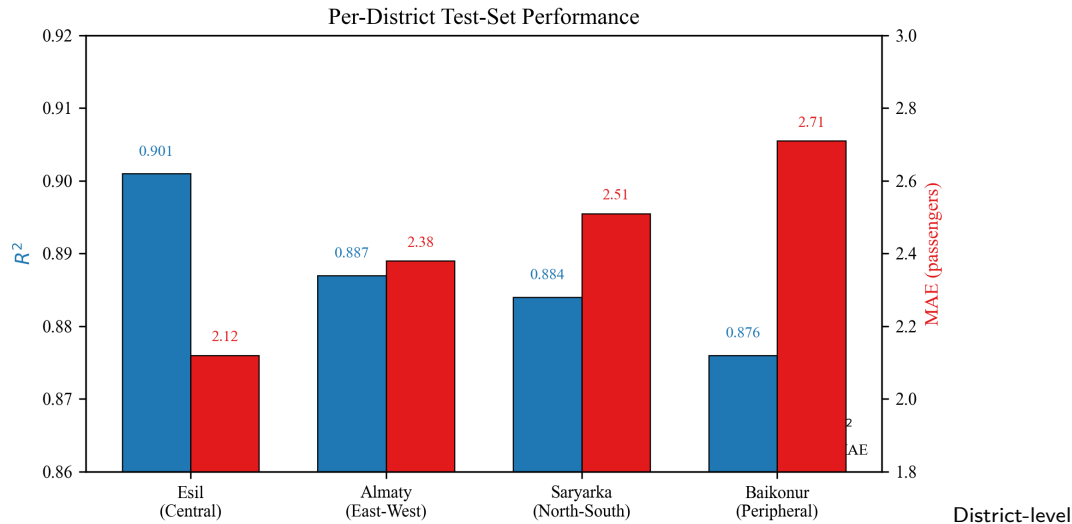
Feature importance

Feature importance

- Weather is the strongest predictor.
- Temporal cyclical encodings capture daily/weekly patterns.
- Graph structure supports spatial transfer effects.



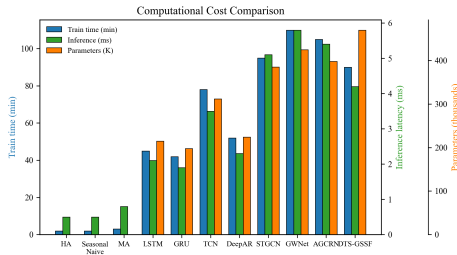
District analysis



results highlight central and peripheral demand patterns.

Operational readiness

- Inference latency: < 5 ms per sample.
- Model size: 542K parameters, 2.2 MB FP32.
- Calibrated NB output supports risk-aware dispatching.
- LoRA adaptation updates only 6% of parameters for route tuning.



Computational cost is suitable for real-time deployment.

Task outcomes

- Confirmed dataset realism and feature expressiveness.
- Developed a new hybrid architecture with adaptive graph learning.
- Demonstrated strong results on synthetic and open datasets.
- Proved the method is operationally viable and generalisable.

Final conclusions

- DTS-GSSF combines temporal, spatial, and uncertainty modules effectively.
- The model is accurate, calibrated, and generalises across cities.
- It supports transit operations with real-time, low-latency predictions.
- Future work: multi-modal transit, dynamic graphs, and edge deployment.